Introduction to Swarm Intelligence

Swarm Intelligence is a new subset of Artificial Intelligence (AI) designed to manage a group of connected machines. We are now entering the age of the Intelligent machines, also called the Internet of Things (IoT), where more and devices are being connected every day.

Swarm intelligence is quickly emerging as a way this connectivity can be harnessed and put to good use.

Swarm intelligence (as the name suggests) comes from mimicking nature. Swarms of social insects, such as ants and bees, operate using a collective intelligence that is greater than any individual member of the swarm. Swarms are therefore highly effective problem-solving groups that can easily deal with the loss of individual members while still completing the task at hand -- a capability that is very desirable for a huge number of applications. Today this concept is being applied in concert with machine learning and distributed computing systems. The result is a group of connected machines that can communicate, coordinate, learn and adapt to reach a specific goal. Check out the video below and its subsequent follow up videos to see how swarm intelligence is applied to a group of drones.

Swarm Intelligence Techniques: Ant Colony Optimization

Ant colony optimization (ACO), introduced by Dorigo in his doctoral dissertation, is a class of optimization algorithms modeled on the actions of an ant colony. ACO is a probabilistic technique useful in problems that deal with finding better paths through graphs. Artificial 'ants'—simulation agents—locate optimal solutions by moving through a parameter space representing all possible solutions. Natural ants lay down pheromones directing each other to resources while exploring their environment. The simulated 'ants' similarly record their positions and the quality of their solutions, so that in later simulation iterations more ants locate for better solutions.

The use of swarm intelligence in telecommunication networks has also been researched, in the form of ant-based routing. This was pioneered separately by Dorigo et al. and Hewlett Packard in the mid-1990s, with a number of variations since. Basically, this uses a probabilistic routing table rewarding/reinforcing the route successfully traversed by each "ant" (a small control packet) which flood the network. Reinforcement of the route in the forwards, reverse direction and both simultaneously has been researched: backwards reinforcement requires a symmetric

DR. VISHAL SHRIVASTAVA, ACET, JAIPUR

network and couples the two directions together; forwards reinforcement rewards a route before the outcome is known (but then one would pay for the cinema before one knows how good the film is). As the system behaves stochastically and is therefore lacking repeatability, there are large hurdles to commercial deployment.

Mobile media and new technologies have the potential to change the threshold for collective action due to swarm intelligence.

The location of transmission infrastructure for wireless communication networks is an important engineering problem involving competing objectives. A minimal selection of locations (or sites) is required subject to providing adequate area coverage for users. A very different-ant inspired swarm intelligence algorithm, stochastic diffusion search (SDS), has been successfully used to provide a general model for this problem, related to circle packing and set covering. It has been shown that the SDS can be applied to identify suitable solutions even for large problem instances.

Particle Swarm Optimization

Particle swarm optimization (PSO) is a global optimization algorithm for dealing with problems in which a best solution can be represented as a point or surface in an n-dimensional space. Hypotheses are plotted in this space and seeded with an initial velocity, as well as a communication channel between the particles. Particles then move through the solution space, and are evaluated according to some fitness criterion after each time step. Over time, particles are accelerated towards those particles within their communication grouping which have better fitness values. The main advantage of such an approach over other global minimization strategies such as simulated annealing is that the large numbers of members that make up the particle swarm make the technique impressively resilient to the problem of local minima.

Nanoparticles are bioengineered particles that can be injected into the body and operate as a system to do things drug treatments cannot. The primary problem with all of our current cancer treatments is most procedures target healthy cells in addition to tumors, causing a whole host of side effects. Nanoparticles by comparison, are custom designed to accumulate ONLY in tumors, while avoiding healthy tissue.

Nanoparticles can be designed to move, sense, and interact with their environment, just like robots. In medicine, we call this embodied intelligence. The challenge thus far has been figuring out how to properly "program" this embodied intelligence to ensure it produces the

Dr. VISHAL SHRIVASTAVA, ACET, JAIPUR

desired outcome.

Swarms are very effective when a group of individual elements (nanoparticles in this case) begin reacting as a group to local information. Swarm intelligence is emerging as the key to which will unlock the true potential of these tiny helpers. Researchers are now reaching out to the gaming community in an effort to crowd source the proper programming for swarm of nanoparticles.

Bee Colony Optimization

The Artificial Bee Colony (ABC) algorithm is a swarm based meta-heuristic algorithm that was introduced by Karaboga in 2005 (Karaboga, 2005) for optimizing numerical problems. It was inspired by the intelligent foraging behavior of honey bees. The algorithm is specifically based on the model proposed by Tereshko and Loengarov (2005) for the foraging behavior of honey bee colonies. The model consists of three essential components: employed and unemployed foraging bees, and food sources. The first two components, employed and unemployed foraging bees, search for rich food sources, which is the third component, close to their hive. The model also defines two leading modes of behavior which are necessary for self-organizing and collective intelligence: recruitment of foragers to rich food sources resulting in positive feedback and abandonment of poor sources by foragers causing negative feedback.

In ABC, a colony of artificial forager bees (agents) search for rich artificial food sources (good solutions for a given problem). To apply ABC, the considered optimization problem is first converted to the problem of finding the best parameter vector which minimizes an objective function. Then, the artificial bees randomly discover a population of initial solution vectors and then iteratively improve them by employing the strategies: moving towards better solutions by means of a neighbor search mechanism while abandoning poor solutions.

The general scheme of the ABC algorithm is as follows:

First Initialization Phase

REPEAT

1. Employed Bees Phase
2. Onlooker Bees Phase
3. Scout Bees Phase
4. Memorize the best solution achieved so far

UNTIL (Cycle=Maximum Cycle Number or a Maximum CPU time)

Dr. VISHAL SHRIVASTAVA, ACET, JAIPUR